Automatic Analysis of Financial Reports

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1 Introduction

1.1 What is Financial Reporting

Financial Reporting involves organizations officially communicating their financial performance and economic status to stakeholders like investors, creditors, and regulators, such as the Australian Securities and Investments Commission (ASIC). These reports are governed by the Australian Accounting Standards (AASB), which align with the International Financial Reporting Standards (IFRS). They offer a standardized business language that translates complex operations into clear, periodic disclosures, including the Financial Position and Statement of Comprehensive Income.

1.2 The Volume and Complexity of Financial Reporting

As of 2026, the financial landscape is defined by an information mystery, in which corporate transparency is at an all-time high, but the data provided is often a drawback due to its format and scale. Current financial and fund statements are highly unstructured and fundamentally heterogeneous. A simple annual report often spans over 150 pages, which includes a complex mix of tables and complicated qualitative narratives. The core of this research is the transition from "passive" data ingestion to "active" intelligence. The industry is moving away from simple extraction tools toward Agentic AI, which can be categorized as autonomous systems that don't just "read" text but use reasoning loops to solve complex classification problems [1]. In this report, the agent is designed to bridge the gap between the unstructured PDF format and the rigid standards used for global financial comparison.

1.3 Problem

The problems that exist can be categorized into two major issues. The first issue is the gap in extracting structured data from unstructured financial reports, resulting in Contextual Decay where extracted data loses its semantic meaning and structure. A solution has been developed to tackle this issue specifically. However, the second issue remains unsolved, in which the data extracted from the first stage needs to be standardized before it can be passed to further analysis. This problem exists because different companies use different labels to describe the same financial concept, making cross-company comparison impossible without standardization. This project is designed to tackle that second problem.

1.4 Report Objective and Scope

The report details the design and implementation of a Domain Mapping Agent that standardizes unstructured financial data extracted from PDFs. While the broader system handles the end-to-end pipeline from ingestion to storage, this research focuses specifically on the Standardization Service. The objective is to demonstrate a three-layered architecture, which includes Knowledge Graphs, Agentic Reasoning, and Deterministic Validation, that can resolve ambiguity and OCR noise [1]. By leveraging GICS for industry contextualization and XBRL for regulatory alignment, this report established a framework for a Verified Ledger [2]. In which this ledger is used as a logically transparent data source.

1.5 Project Significance

This project provides a scalable architecture for Cross-Sector Financial Standardization [3]. By automating the classification of heterogeneous data and aligning it with XBRL/GICS standards, the system:

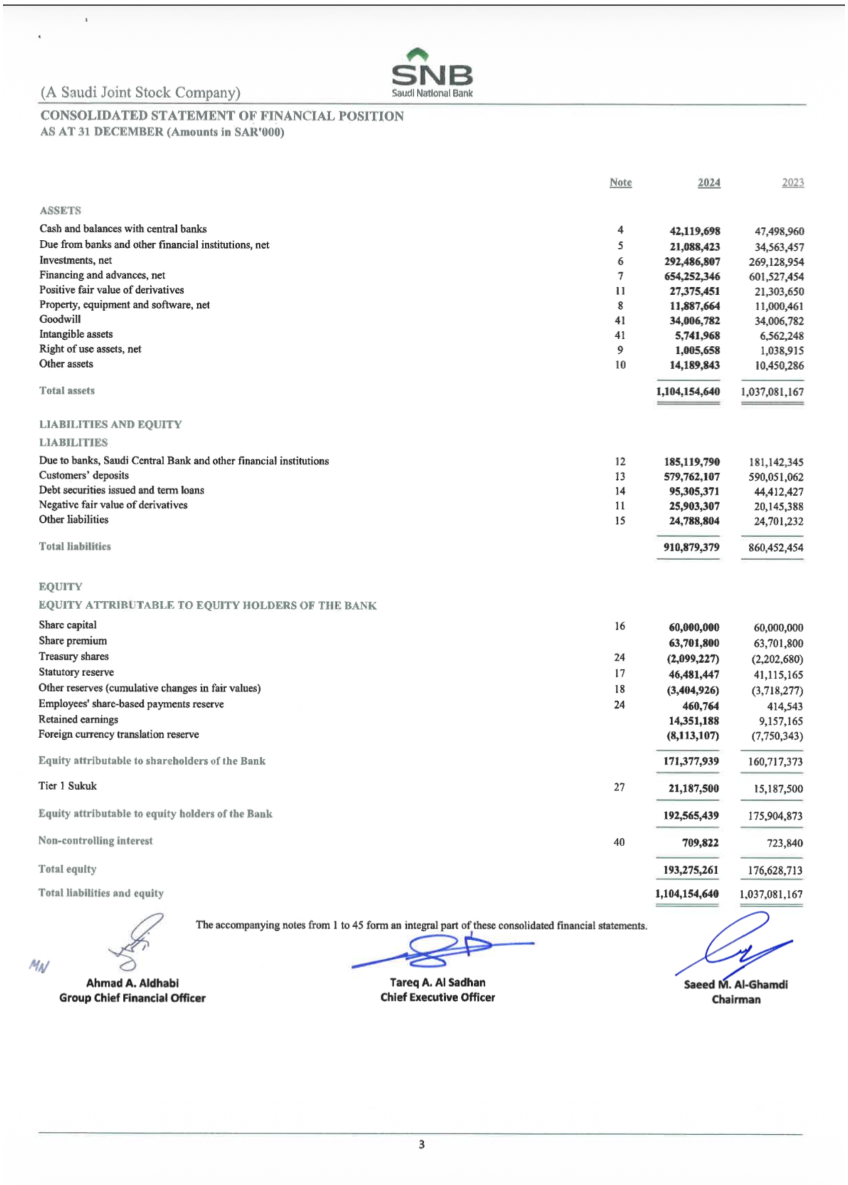
- Reduces Research Latency: Accelerating the "time-to-insight" for analysts [4].
- Enhances Comparability: Enabling "apples-to-apples" comparison across diverse industries [3].
- Ensures Verifiable Intelligence: Moving from probabilistic guesses to logically grounded financial data [1].

2 Literature Survey

2.1 Financial Reporting Concepts

To illustrate the challenge of financial standardization, consider the extract shown in Figure 1. The labels extracted from this report are sector-specific and bear no resemblance to standard XBRL tags.

Without industry context provided by GICS, a system has no principled way to resolve these labels to their correct regulatory equivalents. In order for the system to process financial reports across different companies and sectors, it must first identify the industry context via GICS and then match all field names to their corresponding XBRL tags [3, 2].


(A Saudi Joint Stock Company)

SNB
Saudi National Bank

CONSOLIDATED STATEMENT OF FINANCIAL POSITION
AS AT 31 DECEMBER (Amounts in SAR'000)

	Note	2024	2023
ASSETS			
Cash and balances with central banks	4	**42,119,698**	47,498,960
Due from banks and other financial institutions, net	5	**21,088,423**	34,563,457
Investments, net	6	**292,486,807**	269,128,954
Financing and advances, net	7	**654,252,346**	601,527,454
Positive fair value of derivatives	11	**27,375,451**	21,303,650
Property, equipment and software, net	8	**11,887,664**	11,000,461
Goodwill	41	**34,006,782**	34,006,782
Intangible assets	41	**5,741,968**	6,562,248
Right of use assets, net	9	**1,005,658**	1,038,915
Other assets	10	**14,189,843**	10,450,286
Total assets		**1,104,154,640**	1,037,081,167
LIABILITIES AND EQUITY			
LIABILITIES			
Due to banks, Saudi Central Bank and other financial institutions	12	**185,119,790**	181,142,345
Customers' deposits	13	**579,762,107**	590,051,062
Debt securities issued and term loans	14	**95,305,371**	44,412,427
Negative fair value of derivatives	11	**25,903,307**	20,145,388
Other liabilities	15	**24,788,804**	24,701,232
Total liabilities		**910,879,379**	860,452,454
EQUITY			
EQUITY ATTRIBUTABLE TO EQUITY HOLDERS OF THE BANK			
Share capital	16	**60,000,000**	60,000,000
Share premium		**63,701,800**	63,701,800
Treasury shares	24	**(2,099,227)**	(2,202,680)
Statutory reserve	17	**46,481,447**	41,115,165
Other reserves (cumulative changes in fair values)	18	**(3,404,926)**	(3,718,277)
Employees' share-based payments reserve	24	**460,764**	414,543
Retained earnings		**14,351,188**	9,157,165
Foreign currency translation reserve		**(8,113,107)**	(7,750,343)
Equity attributable to shareholders of the Bank		**171,377,939**	160,717,373
Tier 1 Sukuk	27	**21,187,500**	15,187,500
Equity attributable to equity holders of the Bank		**192,565,439**	175,904,873
Non-controlling interest	40	**709,822**	723,840
Total equity		**193,275,261**	176,628,713
Total liabilities and equity		**1,104,154,640**	1,037,081,167

The accompanying notes from 1 to 45 form an integral part of these consolidated financial statements.

MN

Ahmad A. Aldhabi
Group Chief Financial Officer

Tareq A. Al Sadhan
Chief Executive Officer

Saeed M. Al-Ghamdi
Chairman

3

Figure 1: Extract from Saudi National Bank, showing sector-specific fields that may vary across companies

- XBRL (eXtensible Business Reporting Language): Serving as the technical backbone of the project, which provides a standardized set of machine-readable tags that translates financial items into a regulatory-compliant, system-readable format. Built on a hierarchical taxonomy containing thousand of standardized elements, in which each tag contains not only the identity of a financial fact, but also a structural position, for example `ifrs-full:CurrentAssets` is a child of `ifrs-full:Assets`, meaning the values must mathematically sum to the parent. However, in practice, companies often creating custom XBRL extensions for items not available in the normal taxonomy, and the concept can be validly tagged in different ways depending to the accounting policy choices, making large-scale automated mapping a significant challenge.

- GICS (Global Industry Classification Standard): While XBRL defines the items, GICS provides the structural context of how to interpret them [2]. Launched in 1999 by S&P and MSCI specifically for global investment community, in which GICS classifies companies using a demand-side, market-oriented approach, which groups companies based on their principal business activity and revenue sources rather than the production processes. It has a four-tier hierarchy, consists of 11 sectors, 24 industry groups, 69 industries, and 158 sub-industries. This hierarchy gives the agent a precise ontological map to narrow the XBRL search space, ensuring that only industry-appropriate tags and considered during mapping. However, GICS alone only defines the industry, but not providing guidance of specific accounting treatment or regulatory tag to apply to a given line item.

The limitations of each framework when used in isolation established a challenge the report addresses, which includes accurate automated financial standardization which requires both standards working hand-in-hand with each other. The following sections examine the challenges in applying GICS and XBRL at scale before identifying the gap in existing automated approaches.

2.2 Global Industry Classification Standard (GICS)

2.2.1 Classification Philosophy

The characteristic of the GICS that is different from any other classification is its demand-side, market-oriented approach in categorizing companies. Unlike the Standard Industrial Classification (SIC) system that was established in 1939, which groups companies based on the similarity of their production processes and raw materials, or the North American Industry Classification System (NAICS) which focuses on how companies manufacture rather than what markets they serve, GICS classifies companies according to their principal business activity as determined by the revenue sources and investor perception. The Fama-French algorithm, while widely used in academic research, is simply regrouping of SIC codes into broader statistical buckets and too rough for the fine-grained financial standardization the system requires. GICS was designed specifically for the global investment community and the market-oriented philosophy aligned directly with how usually the financial disclosures are structures and how XBRL tags are assigned.

2.2.2 The Principal Business Activity Rule

The core of the GICS classification is the Principal Business Activity rule, which assigns every company of one GICS code across all four tier of hierarchy through a three-step validation process: Revenue Analysis, followed by Earnings (EBIT) Analysis, and finally Market Perception. This process is critical because GICS classification is not a key-match exercise, but it requires reasoning about the hierarchy of a company's revenue streams in the context of how the investment community perceives the company. A great example for this is a large energy conglomerate may report a lot of revenue from chemical corporations, causing a naive text mining tool to misclassify the company as a chemicals manufacturer. Here, GICS will correctly identify the company's core identity as Integrated Oil and Gas because that is the primary revenue driver and how the community trades it. This gap between keyword frequency and true business identity is precisely the problem the ReAct reasoning loop in this system is designed to solve.

2.2.3 Performance Superiority Over Other Schemes

Research consistently demonstrates that GICS outperforms competing classification schemes in financial research applications. Bhojraj, Lee and Oler (2003) showed that GICS groupings produce lower error rate when predicting company value using comparable companies, and that GICS classified peers are more similar in terms of growth rates, R&D and profit margins, than those identified by SIC or NAICS [3]. Further research by Chan, Lakonishok and Swaminathan (2007) confirmed that GICS is superior in capturing stock return comovements, particularly for large-cap stocks [5]. Boni and Womack (2006) demonstrated that the industry effect captured by GICS is a primary driver of price momentum and investment returns, because analysts focus on specific industries rather than broad production categories [6]. These findings collectively justify the use of GICS as the foundation of the Knowledge Graph in this project.

2.2.4 Challenges and Limitations

Despite all the advantages, GICS has severe review lag problem, although GICS are updated more frequently than that of government systems, formal reclassification typically occurs infrequently, usually annually or a major corporate event, meaning companies that pivot their business model mid-year remain stuck in outdated categories that no longer reflect their true operations. GICS also have a vendor inconsistencies problem, in which financial databases such as Bloomberg, Compustat, and Refinitiv assign different GICS sub-industry codes to the same company based on their own interpretation of revenue segments, which creating classification noise and absence of a universal ground truth. GICS also have a conglomerate dilemma problem, in which the single code requirement forces company that have multiple stream incomes into one bucket, ignoring other operations. Finally, automatic classification introduces semantic ambiguity, companies that use aspirational language are often misclassified into incorrect sector arrangements. These challenges collectively define the problem that we are trying to tackle in this project.

2.3 XBRL and Regulatory Alignment

2.3.1 What XBRL is and How It Works

XBRL (eXtensible Business Reporting Language) is a globally recognized, machine-readable tagging standard for financial data, maintained by XBRL International and mandated by regulators worldwide, including SEC for US companies and ASIC for Australian listed entities under AASB/IFRS alignment. Built on XML, XBRL works by assigning a standardized tag into every financial fact in a report, making each item identifiable and reportable across companies, jurisdictions, and reporting periods. These tags are organized within a hierarchical taxonomy, which the IFRS taxonomy alone contains thousands of standardized elements, where each tag encodes not only the identity of a financial fact but also its structural position relative to the other items. For example `ifrs-full:CurrentAssets` and `ifrs-full:NoncurrentAssets` are children of `ifrs-full:Assets`, meaning their values must be mathematically sum to the parent total. This structure is the foundation of the Deterministic Validation layer of the project, as it provides summation rules that the agent uses to verify whether the classification is consistent.

2.3.2 The Problem of XBRL in Practice

Despite XBRL being a standardization tool, the real-world application produces heterogeneous data, creating challenges for automated financial standardization. Companies frequently create custom XBRL tags that are not covered by the base taxonomy, and the same economic concept can be validly tagged in multiple ways depending on which way the company's accounting policy goes. There are also different companies that may use different label for the same financial concept for their filings, requiring the system to resolve semantic variation rather perform exact matching. OCR noises and PDF extraction errors are also identified as issues regarding the XBRL, as the raw text that is fed to a tagging system is already corrupted before mapping begins, introducing external errors that complicate the mapping. There is also a problem regarding scalability issues, in which this step requires a deep knowledge of IFRS or AASB taxonomy, and cannot be performed at scale with no assist of automation, which is precisely the problem this system is designed to address.

2.3.3 Why XBRL Alone Is Insufficient

While XBRL provides the target that line item must be resolved to, it cannot determine tags are relevant for a given company. A tag like `ifrs-full:InsuranceContractLiabilities` is valid for any company but is only meaningful if the system already knows the company is an insurer, in which applying the tag itself to a mining company will be a classification error that XBRL itself has no mechanism to detect. This is the fundamental limitation: XBRL defines what a financial item is, but there is no way to interpret it within an industry context. Without an external framework to constrain the search space, an automated system faces a taxonomy of thousands of potential tag with no principled way to narrow them given a company. This is where GICS is useful, by identifying the company's industry context first, the agent can prune the XBRL taxonomy tree to only subset of tags that are relevant to the sector, transforming a search problem into a manageable classification task.

2.3.4 XBRL and Knowledge Graphs

Recent research by Li et al. (2021) has explored representing XBRL taxonomies as Knowledge Graphs, where nodes represent individual tags and edges represent the parent-child and sibling relationships between them [7]. The graph representation enables two capabilities that are central to this system. First, there is a capability of a semantic search, rather than performing a keyword lookup through thousands of tags, the agent can traverse the graph to find the most contextually appropriate tag for a given extracted description, guided by the structural relationships already encoded in the taxonomy. Second, it enables summation rule enforcement, the knowledge graph stores which child tags must sum to which parent totals, giving the Deterministic Validation layer the mathematical ground truth it needs to verify every classification decision. Without this graph structure, validation would be impossible, in which the system would have no way of knowing that CurrentAssets + NoncurrentAssets = Assets unless that relationship was explicitly stored in the graph. Together, GICS and XBRL form the combined ground truth layer that underpins the classification and validation architecture of this system.

2.4 Technology Landscape

While GICS and XBRL provide the conceptual ground truth for classification and validation, applying them to unstructured PDF data requires an intelligent reasoning layer capable of handling semantic ambiguity, OCR noise, and hierarchical complexity which are capabilities that simple rule-based systems cannot provide. This section examines the technology landscape that enables such a layer.

2.4.1 Agentic AI and ReAct Framework

The core of this research is centered around Agentic AI, which describes systems that can reach their goals through recursive reasoning loops [1]. While simpler rule-based systems can perform direct tag lookup for well-formatted financial data, they fail fundamentally when encountering ambiguous labels, OCR noise, or conglomerate revenue structures, because they have no mechanism to reason about context, only to match patterns. The Reasoning and Acting (ReAct) framework addresses this limitation by allowing the agent to think through a problem before acting, iterating through Thought-Action-Observation cycles until a confident classification is reached. This is critical for financial standardization because a company's true business identity cannot always be determined from a single line item which requires reasoning across the surrounding document context, the mathematical relationships between fields, and the industry framework simultaneously. The remaining challenge is the "Black Box" effect, where the internal logic of classification becomes difficult for human researchers to audit [1]. This is directly addressed in this system through the Auditability non-functional requirement, which mandates that the agent outputs a transparent thought process for every mapping decision, allowing human verification at each step.

2.4.2 Knowledge Representation and Ontological Mapping

Knowledge Representation is the field of computer science dedicated to structuring information into machine-friendly form through taxonomies and ontologies. While a flat lookup table could theoretically map company names to GICS codes, it cannot encode the parent-child summation rules required for mathematical validation which only a graph structure can simultaneously represent both the classification hierarchy and the arithmetic relationships between nodes [2]. In this research, the Knowledge Graph serves a dual purpose: it acts as the ontological map for GICS-based industry pruning, and as the source of ground truth summation rules for the Deterministic Validation layer. The challenge in this part of the research is classification noise, where different data vendors assign different codes to the same company based on their own interpretations, causing an absence of universal ground truth [8]. The Knowledge Graph addresses this directly by encoding a single authoritative set of GICS-to-XBRL mappings that the agent uses as its sole reference, bypassing vendor inconsistency entirely.

2.4.3 Semantic Computing and Contextual Embeddings

The field is currently moving away from traditional keyword-based searches toward semantic models that prioritize the underlying context of financial language. General-purpose models such as BERT, while powerful, are trained on broad text corpora and lack familiarity with domain-specific financial vocabulary — terms like "unearned premiums", "goodwill impairment", or "net investment in finance lease" carry precise accounting meanings that a general model may misinterpret [4]. FinBERT addresses this by fine-tuning on financial disclosure text, enabling it to correctly distinguish between a company that manufactures oil equipment and one that extracts oil, even when both reports contain identical high-frequency vocabulary. This domain specificity is critical for maintaining GICS precision — a general embedding model would conflate these two companies into the same sector, whereas FinBERT preserves the semantic distinction that the GICS Principal Business Activity rule requires. However, embedding similarity alone cannot guarantee mathematical consistency between classified components and reported totals — a limitation that motivates the Deterministic Validation layer as a necessary second check on top of the Semantic Mapper's output [4].

2.4.4 Multi-Dimensional Classification and Data Integrity

While traditional systems assign a single code identity, modern research is moving towards Multi-Label Classification, which assigns probability vectors across multiple sectors to solve the Conglomerate Dilemma — avoiding the forced assignment of a multi-divisional firm into a single bucket [4, 8]. While this approach better represents the economic reality of diversified firms, it produces probabilistic outputs that are fundamentally incompatible with deterministic validation, in this project it can be seen that a probability vector cannot be used to enforce the summation rule because there is no single authoritative classification to validate against. This system therefore deliberately adopts a single-code GICS assignment, accepting the known limitation of the conglomerate dilemma in exchange for the mathematical guarantees that multi-label approaches cannot provide. This is a conscious design trade-off: verifiability and auditability are prioritized over coverage, on the basis that a deterministically validated single classification is more useful for regulatory financial reporting than a probabilistic multi-label assignment that cannot be independently verified. The ultimate challenge remains the Replicability Crisis, where databases do not always synchronize their GICS updates simultaneously [8], which this system mitigates by maintaining its own authoritative Knowledge Graph rather than relying on external vendor data.

2.5 Software Tools

2.5.1 Natural Language Processing (NLP)

NLP serves as a tool to bridge the gap between unstructured financial narratives and rigid regulatory standards. The problem with keyword matching is the same word can carry a different meaning across industries. For example, "reserves" can be meant by mineral deposit in mining, or even capital in hand for banking. Modern NLP solves this by using contextual embeddings, not just counting words, but understanding the full meaning of a sentence to preserve GICS precision [4, 3]. NLP alone cannot check whether the numbers add up correctly, that is why there is a Deterministic Validation Layer that handles the precision of the mapping.

2.5.2 Machine Learning (ML)

ML is being implemented to resolve the Hierarchical Relationships in the financial tables. The agent will be using Sequence Analysis to determine whether a line item is a subcomponent or a total [4]. The problem with simple heuristics is that formatting is often lost in scanned PDFs, making indentation and visual cues unreliable. ML solves this by learning structural patterns from the data rather than relying on visual formatting [1]. However, ML requires training data, in which the Knowledge Graph provides the ground truth.

2.5.3 Artificial Intelligence (AI)

AI is implemented through the Reasoning and Acting (ReAct) Framework, which serves as the logical controller of the pipeline [1]. The agent is using a dual-branch workflow, which analyzes the company's footprint to assign a GICS sector at the entity level and performs XBRL Mapping for JSON entry [1]. The dual-branch workflow is more favored than a single-branch sequential approach, where GICS classification and XBRL mapping are performed one after another in a fixed sequence, because if GICS is wrong, every subsequent XBRL mapping is wrong with no mechanism for correction. The dual-branch workflow solves this by validating the result in each branch independently before the results are combined. This approach also uses a ReAct loop, which acts as a feedback mechanism that is lacking in the single-branch pipeline.

2.5.4 Vector Database

Vector Databases are used as the Semantic Memory for the AI agent, providing infrastructure to perform non-lexical searches across massive taxonomies [4]. ChromaDB or Pinecone is used to index high-dimensional representations of thousands of potential tags and GICS definitions [4]. Comparing with relational databases such as PostgreSQL, Vector Databases have the advantage of finding semantically similar tags even when the input is corrupted, while relational databases can only perform pattern-based lookups. When the agent gets a JSON entry, it performs a Nearest Neighbor Search to find the tags with the highest similarity. Vector Databases also allow for Metadata Filtering, ensuring only industrially relevant tags are included in the search. However, Vector Databases only return the most similar match, not necessarily the correct one, which is why results are always passed through the Validation Layer.

2.5.5 Knowledge Graph

Knowledge Graph is the backbone of the project, representing the ground truth of the system. It stores information as a network of Nodes and Edges, where each node represents a GICS code or XBRL tag and each edge represents the relationship between them [2]. Compared to relational databases, which can store parent-child relationships but cannot efficiently perform graph traversal for contextual pruning across thousands of nodes, the Knowledge Graph is the only structure that simultaneously combines ontological mapping, hierarchical relationship traversal, and summation rule enforcement [7]. This is essential for resolving Contextual Ambiguity — rather than searching for a text match alone, the Knowledge Graph also validates mathematically, acting as a bridge between unstructured financial data and regulatory compliance [2].

2.6 Integrating the Technology Landscape with Reporting Concepts

As mentioned in section 1.3, there are 2 problems that are stated, in which those are the extraction of the table from the financial report, and the standardization of the data pulled from the financial report by itself. There are prototypes that tries to resolve the extraction of the data from the financial report, but there are a much harder issue that lies into the second problem. It will include the usage of multiple technologies related to interpreting semantic meanings of the fields.

3 Solution Design

3.1 Features

The solution focuses on scanning financial reports and standardising them according to two international standards: GICS for company classification and IFRS XBRL for financial field tagging. This standardisation enables direct comparison between different companies regardless of how they present their financial data. The system also includes a validation layer to handle fields that cannot be automatically mapped. This ensures that any unrecognised or ambiguous line items are flagged for review rather than silently ignored, maintaining the integrity of the output ledger.

3.2 Use Case

This section is describing the use case of the Domain Mapping Agent, in which there are 2 perspective, such as Administrator and User. The administrator mainly oversees the Knowledge Graph, while the user interacts with the system to standardize and retrieve data.

UC1: Update Knowledge Graph via File Upload

When a new XBRL standard or GICS classification is released, the Administrator uploads the latest taxonomy file through the system interface. The system parses the uploaded file, extracts the updated nodes and relationships, and updates the Knowledge Graph and Vector Database accordingly. This ensures the agent always operates on the most current XBRL and GICS versions, keeping all mappings up to date. For example, the user will upload the brand new XBRL and GICS versions, either as an XML file or as an Python file, and the system will return successful response if completed.

UC2: Getting GICS Sector and XBRL tags from GICS

When the user wants to retrieve the GICS sector and initial XBRL tags for a given company, the user submits a sample API response from the extracted data, containing field names and values, via the Knowledge Layer endpoint. The system queries the Knowledge Graph, identifies the GICS sub-industry, and returns the relevant XBRL tags for that sector. This allows the user to verify that the Knowledge Layer is correctly identifying the industry context and narrowing the search space to only industry-relevant XBRL tags before proceeding to the next stage. For example, the user will upload a JSON full of fields and values, and then the system will return the GICS sector and the fields that are followed

UC3: Getting Initial XBRL Mapping

When the user wants to retrieve the initial XBRL mapping candidates, the user submits a sample API response from the extracted data, containing field names and values, via the Reasoning Layer endpoint. The system first performs the steps described in UC2 to identify the GICS sector and retrieve relevant XBRL tags. The Semantic Mapper then converts the extracted labels into vectors and queries the Vector Database for semantic matching [4], while the ReAct Agent evaluates the candidate tags through the Thought-Action-Observation-Resolution cycle [1]. The endpoint returns each extracted field label alongside its candidate XBRL tag assignment. For example, the user will upload a JSON full of fields and values, and then the system will return the XBRL fields that are matched with the fields that are existed, and note that there may be empty fields.

UC4: Validation Layer Check

When the user wants to check whether the tags match the field names, it will call the Validation Layer. The user will submit the result from UC3, calling the Validation Layer endpoint. The system will perform several checks, such as the Structural Contextualizer will check the hierarchy, and also there are the sum validation checks as well, if there is a mistake it will send a rejection signal which in the big picture is going back to the Reasoning Layer. However, here will be returned as a JSON response, as for the successful responses, will be returned as is. For example, the user will upload a JSON full of fields and values, and then the system will return whether it is correct or not, since it is only the validation layer checks.

UC5: Full Pipeline Test

When the user wants to check the full standardization pipeline, the user uploads the extracted data containing field names and values via the full pipeline endpoint. The system will run UC2, UC3, and UC4 as a unit, which contains all of the identification of GICS sector, mapping the XBRL tags and verifying all the tags based on the sum. It will return as a verified result as an Excel file or a JSON response, configureable in the endpoint. The user may upload the document, the same as in Figure 1, and then it will output all of the fields and values based on the XBRL and GICS.

3.3 Architecture

The proposed solution is divided into 3 main layers, in which there are Knowledge Layer, Reasoning Layer and the Verification Layer. Knowledge Layer will be consisting the Vector Databases and the Knowledge Graph of the system, in which the Knowledge Graph will be storing the nodes and edges of GICS and XBRL, and the relationships between them, while Vector Database will store the semantic meanings of the XBRL tags in order for making the other layers easier to find the semantic meaning of a certain XBRL tag. There is also a Reasoning Layer, in which it has a mechanism of the GICS pruned data in which it only puts out the XBRL tags related to the GICS, the ReAct agent to execute the procedure of finding which tags to investigate and assign, and the Semantic Mapper which, after the ReAct Agent identifies candidate tags, queries the Vector Database to find the closest matching XBRL tag using cosine similarity. There is also the Validation Layer, in which there are Structural Contextualizer to clean up the structures of the produced tags from the ReAct agent, and there is also a Summation Check to check whether all the tags are checked, if there is something wrong in this check, it will be sent back to the Reasoning Layer. The Validation Layer will be sending out the Ledger produced as a JSON API endpoint or an Excel Document. Unlike existing approaches such as FinBERT and rule-based systems which either classify without validating or tag without handling unstructured input, this system combines all three layers into a single deterministic pipeline. The solution is divided into several key concepts:

- Contextual Perception: The agent identifies the company's GICS sector to "prune" the XBRL taxonomy tree, narrowing the search space to industry-appropriate tags [2, 3].
- Reasoning and Acting (ReAct) Loop: The agent iterates through cycles of internal reasoning and taxonomy queries [1]. It "thinks" about the sector context, "acts" by querying specific XBRL definitions, and "observes" whether the extracted text satisfies the accounting criteria.
- Ontological Mapping: Instead of simple similarity, the agent uses Chain-of-Thought (CoT) reasoning to ensure that a value like "unearned premiums" is mapped specifically to the insurance-sector XBRL tag rather than a general liability [4].

- Deterministic Validation: The agent treats the classification as a mathematical identity, enforcing the summation rules from the Knowledge Graph to ensure every output is financially consistent:

$$Total\ Reported = \Sigma(Component\ Extracted)$$

If the sum of the components extracted from the text does not match the reported totals, the agent autonomously re-evaluates the classification of the sub-variables.

The figure below illustrates the proposed system architecture, spanning across the first and second half of the project. The system is divided into three main services: the Input Service, the Extraction Service, and the Standardization Service.

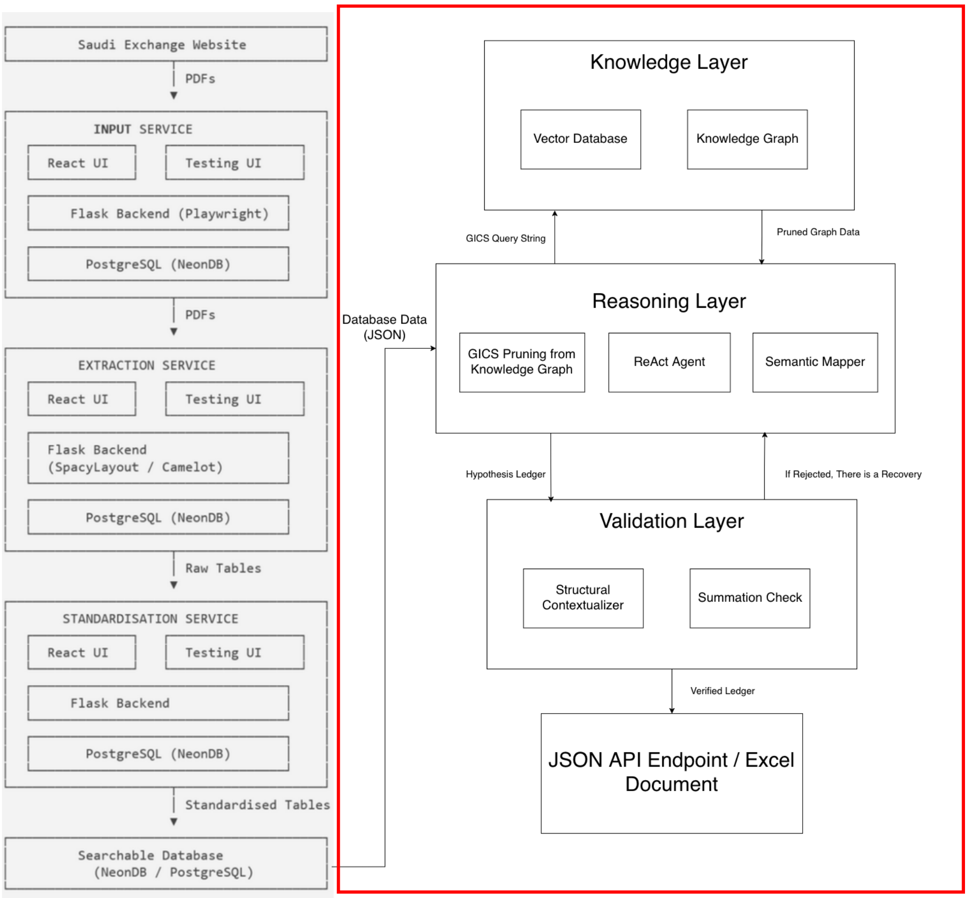


Figure 2: Proposed System Architecture featuring the Agentic Standardization Service (Red Box)

The area which are boxed in the red box is the focus of this report. This includes the AI layer in which the data will be categorized based on the proper entity and tags. While the framework is applicable everywhere, this report demonstrates the capabilities using the data from the Saudi Exchange [9].

3.4 Knowledge Graph Design

3.4.1 Competency Questions

The Knowledge Graph is designed to answer these questions:

Table 1: Knowledge Graph Competency Questions

Question	Query	Example Answer
What XBRL tags correspond to a particular sector?	`SELECT ?tagName WHERE { gics:40 kg:PARENT_OF+ ?si . ?si kg:HAS_RELEVANT_TAG ?t . ?t ifrs:tagName ?tagName }`	LoansAndAdvancesToCustomers, DepositsFromCustomers (461 total)
What GICS sub-industry does a company belong to?	`SELECT ?siName WHERE { gics:40 kg:PARENT_OF+ ?si . ?si rdfs:label ?siName }`	40101015 — Regional Banks
What are the child tags of a given XBRL parent tag?	`SELECT ?child WHERE { ifrs:Assets kg:HAS_CHILD_TAG ?t . ?t ifrs:tagName ?child }`	CurrentAssets, NoncurrentAssets, Liabilities, Equity
What is the summation rule for a certain XBRL tag?	`SELECT ?child WHERE { ifrs:Equity kg:HAS_CHILD_TAG ?t . ?t ifrs:tagName ?child }`	IssuedCapital + SharePremium + RetainedEarnings + NoncontrollingInterests
Which XBRL tags are unique to a specific sub-industry?	`SELECT ?tagName WHERE { gics:40101015 kg:HAS_RELEVANT_TAG ?t . ?t ifrs:isUniversal "false"^^xsd:boolean }`	LoansAndAdvancesToCustomers, CashAndBalancesWithCentralBank

The questions justify the usage of the Knowledge Graph over a normal relational database, since questions 3, 4 and 5 are hierarchical structure questions that needed a graph search.

3.4.2 Knowledge Graph Classifications

The Knowledge Graph consists of the two main categories of nodes: GICS nodes which included the four-tier hierarchy and the XBRL nodes representing the tags. The table below will outline the node types and descriptions.

Table 2: Knowledge Graph Node Classifications

Node Type	Description	Example
GICS Sector	The company's sector, the top layer of GICS	Financials
GICS Industry Group	The company's industry groups	Banks
GICS Industry	The company's industry	Banks
GICS Sub-Industry	The company's sub-industry, which is the last layer of GICS, connects to XBRL tags	Commercial Banks
XBRL Tag	Standardized financial field from XBRL taxonomy	`ifrs-full:Revenue`

3.4.3 Knowledge Graph Relationships

The relationships between nodes are between the GICS hierarchy, the XBRL mappings, and the summation rules. The table below represents better of what the rules are.

3.4.4 Knowledge Graph Diagram Structure

Here is the proposed Knowledge Graph Diagram Structure.

Table 3: Knowledge Graph Relationships

Relationship	From	To	Description
PARENT_OF	GICS Sector	GICS Industry Group	Downward hierarchy from Sector to Industry Group
PARENT_OF	GICS Industry Group	GICS Industry	Downward hierarchy from Industry Group to Industry
PARENT_OF	GICS Industry	GICS Sub-Industry	Downward hierarchy from Industry to Sub-Industry
HAS_RELEVANT_TAG	GICS Sub-Industry	XBRL Tag	Links XBRL tags to their corresponding Sub-Industry
HAS_CHILD_TAG	XBRL Tag (parent)	XBRL Tag (child)	Encodes parent-child hierarchical rules between XBRL tags

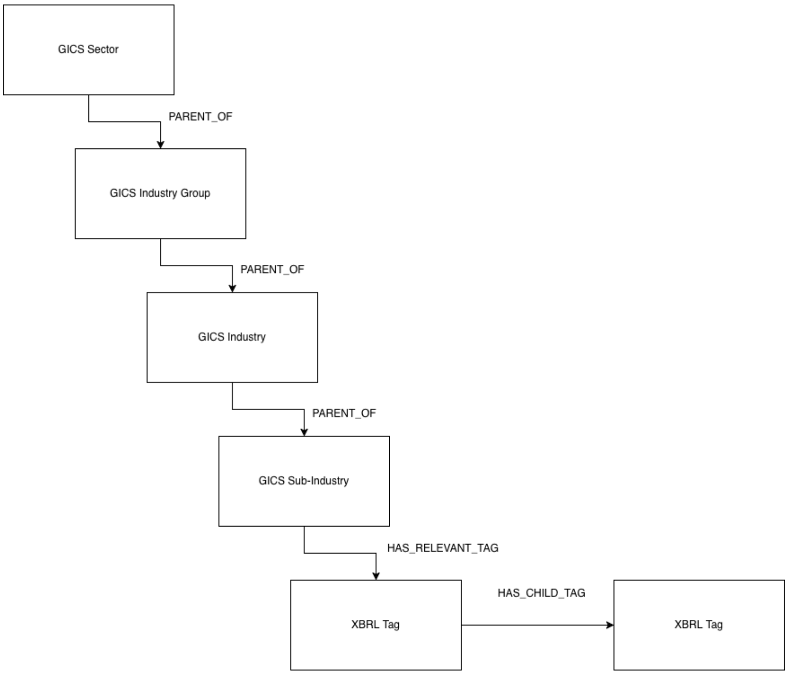


Figure 3: Knowledge Graph Structure Diagram

3.5 Implementation of the Domain Mapping Agent

The Domain Mapping Agent serves as the core of a standardization service, transforming the PDF into a structured financial ledger. The agent provides a suite of capabilities designed to overcome the structural opacity of PDFs and the limitations of manual normalization [3]. The features of the agent is as follows:

- Align the extracted data with an identified GICS category.
- Retrieve the relevant XBRL tags based on the identified GICS category.
- Find and map the extracted labels to their closest matching XBRL tags based on their meaning.
- Verify the parent-child structure of the mapped data.
- Check the completeness of the mapping, if there are mistakes or ambiguous mappings, the data is passed back to the agent for re-evaluation.
- Export the verified mapping as a JSON API endpoint or Excel Document.

In Figure 2, the Domain Mapping Agent is the combination between the Knowledge Layer, the Reasoning Layer and the Validation Layer. The Domain Mapping agent will put out a result either as a REST API JSON or as an Excel File based on the mapping.

3.5.1 Knowledge Layer

The first task of the agent is to define the boundaries of the search space. To prevent pathfinding errors, which can cause a different sector's asset to be mapped to another, the agent performs Knowledge Contextualization [2].

- Identity Mapping: the agent uses the company name as a key to the Knowledge Graph, which contains entities to their respective GICS [2].
- Contextual Pruning: Once the industry is identified, the agent searches the Knowledge Graph and disables the nodes that are not related [4].
- Constraint Loading: The domain is narrowed so that the one field that is unique to the industry is highlighted [3].

3.5.2 Agentic Reasoning (ReAct Agent)

When the agent encounters an extracted variable with low semantic confidence, it initiates the Thought-Action-Observe-Resolution cycle to resolve the ambiguity [1].

- Thought: Upon encountering an extracted variable with low semantic confidence, the agent initializes a reasoning cycle to assess the context of the surrounding document.
- Action: The agent executes a spatial analysis of contiguous data cells and neighboring rows to identify a "semantic anchor", a nearby field with a high-confidence XBRL mapping that can clarify the unknown variable's role.
- Observation: The agent calculates the mathematical delta between the identified sub-variables and the reported totals, determining if the current classification satisfies the summation rules defined in the Knowledge Layer.
- Resolution: By reconciling the structural position of the variable with its mathematical contribution to the ledger, the agent resolves the ambiguity and updates the standardized entry.

3.5.3 Semantic Fact-Alignment (Semantic Mapper)

The Semantic Mapper serves as the noise-resilience layer of the agent, responsible for resolving the gap between raw extracted descriptions and clean Knowledge Graph nodes even in the presence of OCR errors and typographical mistakes. Rather than relying on exact string matching, the Semantic Mapper uses lightweight contextual encoders to generate high-dimensional embeddings for both the extracted description and the candidate XBRL tags, measuring alignment using cosine similarity to ensure reliable matching even when labels are corrupted.

- Lightweight encoders: Encoders are used to generate high-dimensional embeddings for both description and XBRL models [4].
- Similarity Metric: Alignments are determined using cosine similarity [4].
- Mapping: Even with errors, this approach ensures there is a clean node.

3.5.4 Validation and Logic Recovery Loop

The hypothesis ledger will later be sent to the Validation Layer and trigger a feedback loop to ensure things are going correctly. There are several steps, as such.

- Integrity Checks, in which the ledger will undergo a Summation Check for the total amount matching and make sure the structure of the report is correct.
- Rejection Signal: If there is a failure, it will go back to the Reasoning Layer to do a Logic Recovery.
- Autonomous Correction: The agent treats the rejection as a new observation, re-evaluating the reasoning path, and queries the Knowledge Layer for the best candidate nodes [1]. This will be repeated until the correct items are produced.

3.6 Testing and Evaluation

To assess the effectiveness of this mapping agent, this section provides the metrics that will be used to measure the performance of each layer of the three-layered architecture. Each metric corresponds directly to a specific component of a system, ensuring both the system are targeted and comprehensive.

3.6.1 GICS Classification Accuracy

The first metric measures whether the agent correctly identifies the company's GICS sector during the Knowledge Layer phase, before any XBRL pruning takes place. The metric is foundational because an incorrect GICS assignment propagates errors through every subsequent steps, making the whole mapping agent run incorrectly. The benchmark for this evaluation will be a manually verified reference set of GICS classifications for a representative sample of Saudi Exchange companies, cross-referenced against Bloomberg and MSCI classification data to establish ground truth.

$$\text{GICS Accuracy} = \frac{\text{Correct GICS Assignments}}{\text{Total Companies Processed}} \times 100\% \tag{1}$$

3.6.2 XBRL Mapping Accuracy

The second metric measures whether the mapper correctly resolves extracted financial line to their XBRL tags. A mapping is considered correct if the tag matches the expected IFRS taxonomy tag for the line tiem, as determined by manual verification against a reference set. This metric directly evaluates the cosine similarity alignment approach described in Section 3.5.3, and its robustness against OCR noise and typographical errors in the source PDF.

$$\text{XBRL Mapping Accuracy} = \frac{\text{Correctly Mapped Line Items}}{\text{Total Line Items Processed}} \times 100\% \tag{2}$$

3.6.3 Deterministic Validation Pass Rate

The third metric measures how often the agent's classified components satisfy the summation rule on the first pass through the Validation Layer, without the need of Logic Recovery cycle. A high pass rate indicates that the initial GICS pruning and XBRL mapping are producing consistent results, while low pass rate shows that the ReAct loop is carrying a heavy burden of correction. This is useful that it can detect whether errors are systemic in the classification logic or isolated for specific cases.

$$\text{Validation Pass Rate} = \frac{\text{Ledgers Passing Summation Check}}{\text{Total Ledgers Processed}} \times 100\% \tag{3}$$

3.6.4 Logic Recovery Rate

The fourth metric measures how often the agent successfully corrects a failed validation through the ReAct self-correction loop. This metric evaluates the agentic reasoning capability of the system, a high recovery rate shows that the Thought-Action-Observation cycle is effective at resolving any ambiguities in classification, while a low recovery rate would indicate that the agent is encountering classification cases beyond its current reasoning capability.

$$\text{Logic Recovery Rate} = \frac{\text{Successfully Recovered Ledgers}}{\text{Total Rejected Ledgers}} \times 100 \tag{4}$$

3.6.5 Overall Accuracy

The final metric is the combination of the three core performance measures into a single aggregated score, providing an overall assessment of system quality. This provides a single figure that summarizes the whole reliability across classification, mapping and validation. A minimum target of 85% overall accuracy is set as the benchmark for a successful implementation, consistent with the accuracy targets established for similar financial standardization pipelines [1].

$$\text{Overall Accuracy} = \frac{\text{GICS Accuracy} + \text{XBRL Mapping Accuracy} + \text{Validation Pass Rate}}{3} \tag{5}$$

4 Project Plan

4.1 Research Project A Review

During project A, an extensive literature survey is conducted covering the two foundational frameworks that underpin the system. For GICS, the survey examine the classification philosophy, the Principal Business Activity rule, performance superiority over competing schemes such as SIC and NAICS, and the known challenges that came within the domain. For XBRL, the survey covers the taxonomy structure, real-world heterogeneity problems, the insufficiency of XBRL alone without industry context, and the role of Knowledge Graph in enabling semantic search and summation rule. The technology landscape was also critically examined, comparing the ReAct framework against rule-based alternatives, justifying FinBERT over general-purpose BERT for financial classification, and establishing Vector Databases and Knowledge Graphs as the appropriate data infrastructure. A research gap was identified between existing classification approaches and deterministic regulatory alignment, which no current tool addresses within a single automated pipeline.

Based on these findings, the Domain Mapping Agent was designed with a three-layered architecture consisting of a Knowledge Layer for GICS-aware pruning, a Reasoning Layer for ReAct-based agentic classification, and a Validation Layer for deterministic mathematical verification. Functional and non-functional requirements were established, and evaluation metrics were defined against a minimum target of 85% overall accuracy.

Beyond the design work, the GICS-to-XBRL Knowledge Graph was successfully constructed during Research Project A, providing a verified ground truth layer that the remaining implementation phases can build directly upon. However, the Vector Database as the semantic search layer for navigating the Knowledge Graph has not yet been implemented. The following components therefore remain to be completed in Research Project B: the Vector Database, the ReAct Agent, the Semantic Mapper, the Validation Layer, and full evaluation against the metrics defined in Section 3.6.

4.2 Plans for Research Project B

Research Project B will focus on the full implementation and evaluation of the Domain Mapping Agent, building on the Knowledge Graph and solution design established in Research Project A.

- Phase 1 — Vector Database Setup (Weeks 1–2): The Vector Database will be constructed and populated with high-dimensional embeddings for all GICS definitions and XBRL tags stored in the Knowledge Graph. This establishes the semantic search layer that allows the agent to navigate the Knowledge Graph through nearest-neighbor queries rather than exact string matching. ChromaDB or Pinecone will be evaluated and selected based on performance against the Saudi Exchange dataset.

- Phase 2 — ReAct Agent Implementation (Weeks 3–5): The Thought-Action-Observation-Resolution cycle will be implemented and integrated with the Knowledge Graph and Vector Database. The agent will be tested against a sample of Saudi Exchange companies to verify that GICS sector identification and XBRL taxonomy pruning are functioning correctly before proceeding to the next phase.

- Phase 3 — Semantic Mapper and Validation Layer (Weeks 6–7): The FinBERT-based Semantic Mapper will be implemented to perform cosine similarity alignment between extracted line item descriptions and candidate XBRL tags. The Deterministic Validation Layer will then be integrated to enforce the summation rule *Total Reported* = Σ*(Component Extracted)* on every classification output, triggering the Logic Recovery loop when validation fails.

- Phase 4 — Integration and Evaluation (Weeks 8–9): All three layers will be integrated into a single end-to-end pipeline and evaluated against the five metrics defined in Section 3.6. Results will be measured against the minimum target of 85% overall accuracy. Where metrics fall short of the target, the agent will be iteratively refined until the threshold is met or the limitation is documented as a finding for future work.

- Phase 5 — Report Writing and Submission (Week 10): The final research report will be written documenting the implementation decisions, evaluation results, and findings from Research Project B. This includes a discussion of metrics achieved against the defined targets, any limitations encountered during implementation, and recommendations for future improvements to the pipeline.

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